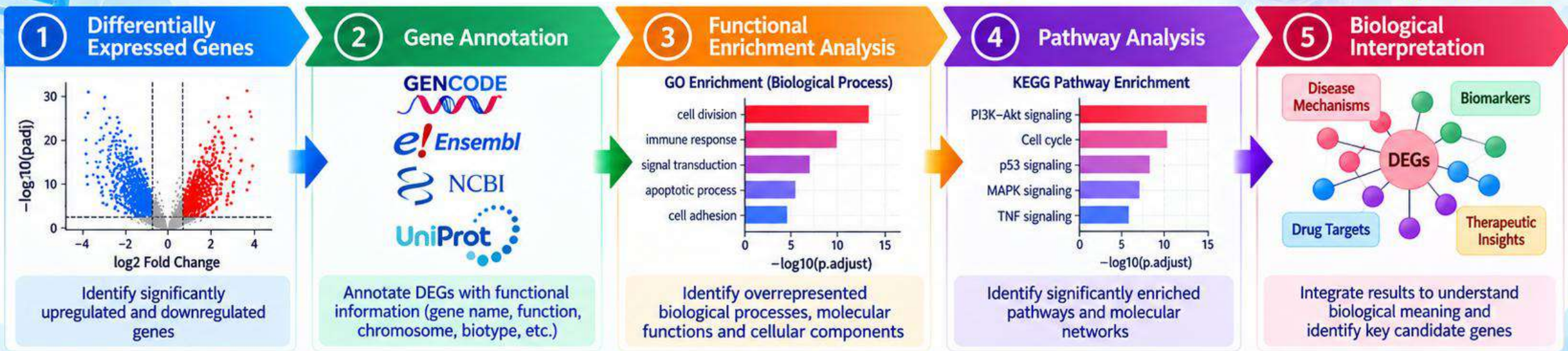


LEARNING OBJECTIVES

By the end of this session, participants will be able to:

1



Understand

the concept and importance of gene annotation after DEG analysis.

2



Identify

different gene identifiers and map them to gene symbols and functional information.

3



Perform

functional annotation of differentially expressed genes using Gene Ontology (GO).

4



Analyze

enriched biological pathways using KEGG, Reactome, and other pathway databases.

5



Interpret

enrichment results to connect DEGs with biological processes and molecular functions.

6



Visualize

functional and pathway enrichment results using publication-quality plots.

7



Explore

protein-protein interaction (PPI) networks and identify potential hub genes.

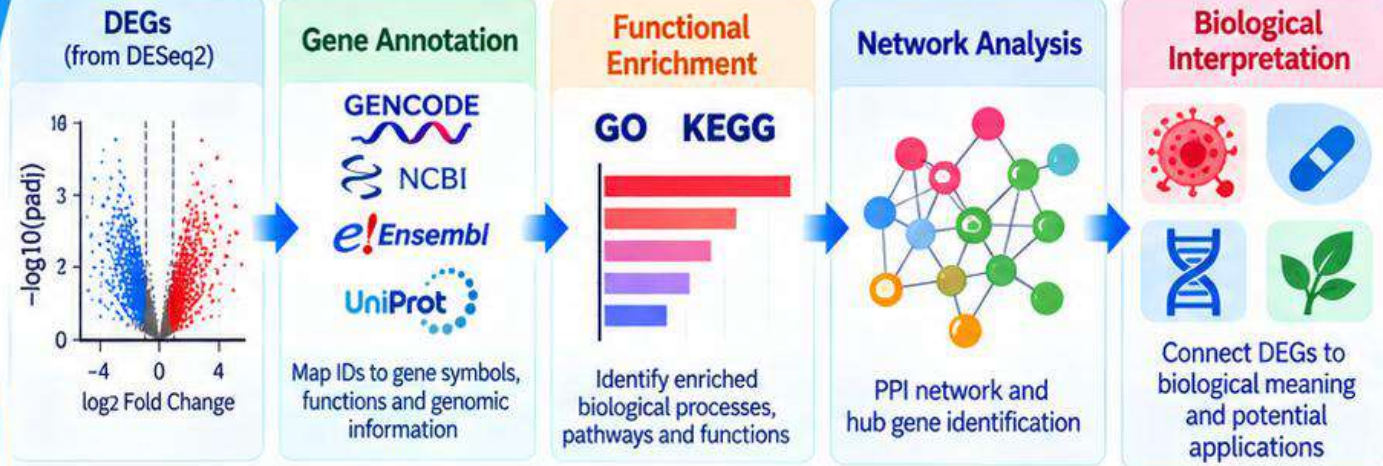
8



Integrate

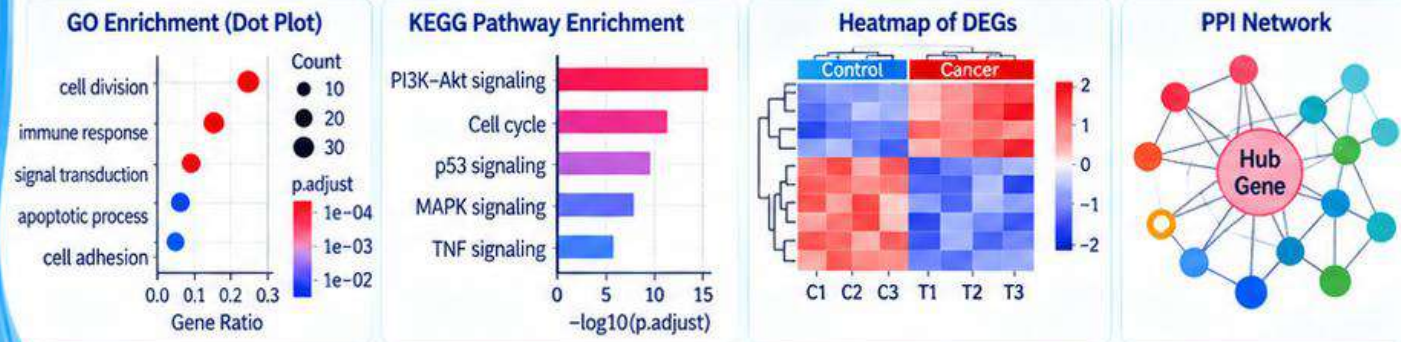
DEG, functional, and pathway information to derive biological insights.

From DEGs to Biological Insights



Genes in Context for a Healthier Tomorrow

Example Outputs



Genes
Functions
Pathways



Data
Knowledge
Biology



Insights
Better Health
Tomorrow

From Data to Discovery for a Healthier Tomorrow!

3 What is Gene Annotation?

Gene annotation = assigning biological information to a gene

It provides information about:



Gene identity
(e.g., Ensembl ID, Entrez ID)

ABC

Gene symbol / name
(e.g., TP53)



Gene function
(what the gene does)



Protein product
(protein sequence, structure)



Cellular location
(e.g., nucleus, cytoplasm)



Biological processes
(e.g., cell cycle, apoptosis)



Molecular functions
(e.g., DNA binding, kinase activity)



Pathways
(e.g., KEGG, Reactome)



Protein domains
(e.g., Pfam, InterPro)



Disease associations
(e.g., cancer, genetic disorders)

Simple concept

Gene ID
(ENSG, Entrez, etc.)

ENSG00000141510

Gene identity
(symbol/name)

TP53

Biological function

Tumor suppressor

Pathway & disease link

Cell cycle, p53 pathway

*A Gene
A Function
A Bigger Story!*

4 Why Do We Need Gene Annotation?

After DEG analysis, we may obtain:

List of DEGs (Gene IDs)

ENSG00000141510
ENSG00000146648
ENSG00000171862
ENSG00000155657
ENSG00000198727
ENSG00000204287
ENSG00000130203
ENSG00000157764
...



These IDs alone do not tell us much about their biological roles.

From IDs to Insights!

Annotation converts

Annotated Information

Gene ID	Gene Symbol	Function / Description
ENSG00000141510	TP53	Tumor suppressor protein
ENSG00000146648	BRCA1	DNA repair protein
ENSG00000171862	EGFR	Receptor tyrosine kinase
ENSG00000155657	MYC	Transcription factor
...	...	...

DEG analysis tells us **which genes changed**, while annotation tells us **what those genes are and what they do**.



DEGs
(Which genes?)



Annotation
(What are they?)



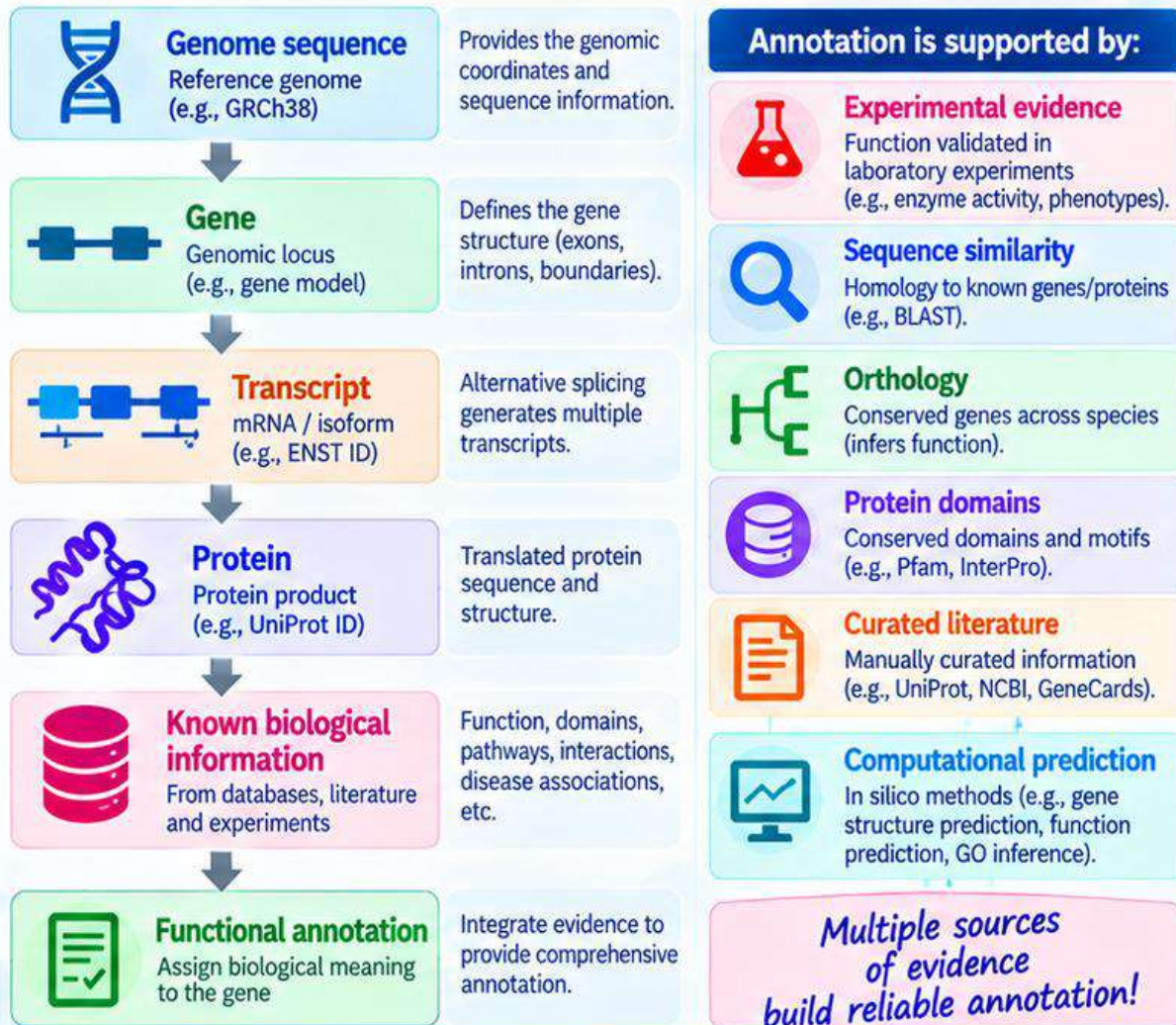
Biological Insights
(Why it matters?)

*Annotate Today
Understand Tomorrow*



5 Basis of Gene Annotation

Gene annotation is based on reference databases and biological evidence.



From Sequence to Function

Reference Databases + Biological Evidence = Functional Annotation

6 Different Types of Annotation

Different annotation types provide complementary information about a gene and its products.

Annotation type	Information obtained	Example	Database / Tool
Gene annotation	Gene identity (gene symbol, name, chromosome, location)	TP53 chr17:7,566,964-7,590,856	GENCODE NCBI
Transcript annotation	Transcript/isoform information (exon-intron structure, isoforms)	ENST00000269305	GENCODE e!Ensembl
Protein annotation	Protein function (protein sequence, structure, name)	P04637 (tumor protein p53)	UniProt
GO annotation	Biological function (BP, MF, CC)	GO:0006915 (apoptotic process)	Gene Ontology
Pathway annotation	Biological pathways	hsa04115 (p53 signaling pathway)	KEGG Reactome
Domain annotation	Protein domains and family information	PF00870 (p53 DNA-binding domain)	Pfam InterPro
Disease annotation	Disease associations (e.g., cancers, genetic disorders)	Breast cancer, Li-Fraumeni syndrome	DisGeNET GeneCards

Integrating multiple annotation types provides a comprehensive view of gene function and its role in biology.

More Knowledge
Deeper Insights!



8 Major Annotation Databases

Multiple Databases
A Complete View

Different databases provide complementary information about genes and proteins



- Gene and transcript information
- Genomic coordinates
- Gene identifiers (ENSG, ENST, etc.)
- Comparative genomics

Genome annotation and gene models



- Gene information
- Gene IDs
- RefSeq sequences
- Genomic information
- Link to other databases

Comprehensive genetic information



- Protein information
- Protein function
- Protein domains
- Subcellular localization
- Post-translational modifications
- Interactions
- Disease relevance

Protein functional annotation



Gene Ontology

- Biological processes (BP)
- Molecular functions (MF)
- Cellular components (CC)
- Controlled vocabulary
- Hierarchical structure

Functional description of genes



- Biological pathways
- Molecular interactions
- Metabolic pathways
- Signaling pathways
- Disease pathways

Pathway and network information

From Gene IDs to Biological Meaning

10 Gene Ontology (GO)

A Common Language
for Gene Function

Gene Ontology (GO) provides standardized descriptions of gene function.



- ✓ Controlled vocabulary
- ✓ Hierarchical structure
- ✓ Used across species
- ✓ Describes gene products

Three major categories

Biological Process (BP)

What biological process does the gene participate in?



Examples:

- Cell cycle
- Apoptosis
- DNA repair
- Immune response

Molecular Function (MF)

What does the gene/protein do?



Examples:

- DNA binding
- ATP binding
- Protein kinase activity
- Transcription factor activity

Cellular Component (CC)

Where does the gene/protein act?



Examples:

- Nucleus
- Mitochondrion
- Plasma membrane
- Cytoplasm

9 UniProt Annotation

Proteins
Give Life to Genes

What can UniProt tell us?

A gene/protein can be associated with:



Protein name



Post-translational modifications



Function



Interactions



Gene name



Disease relevance



Subcellular location



Sequence information

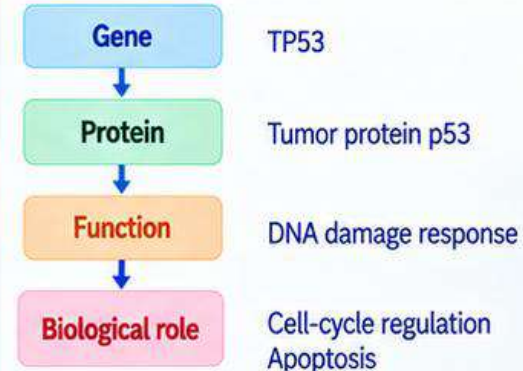


Protein domains



Cross-references (to other databases)

Example: TP53 in UniProt



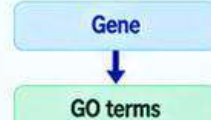
UniProt provides detailed protein-level information that helps understand gene function.

11 GO Annotation vs GO Enrichment

These are different concepts.

GO Annotation

Assigns GO terms to a gene.



Example: TP53

- DNA binding (MF)
- Apoptotic process (BP)
- Nucleus (CC)

Describes the function of individual genes.

GO Enrichment

Identifies overrepresented GO terms in a list of genes (e.g., DEGs).



Example: DEGs

- Cell cycle ↑↑
- DNA replication ↑
- DNA repair ↑

Finds common biological themes in a gene set.

12 Why Perform Functional Enrichment?

From
Gene Lists
to Big Insights

A DEG list may contain hundreds or thousands of genes.

Looking at every gene individually is difficult.

Enrichment analysis asks:
"Are particular biological functions or pathways represented more than expected in my DEG list?"

This allows us to move from:

Individual genes
→ Biological processes
→ Biological pathways

Data
to
Biology

Gene Annotation of Differentially Expressed Genes

From Differential Expression to Biological Insights

RNA-seq Analysis Workshop
Bioinformatics & Computational Biology

2 Overall Workflow

DEG Dataset
↓
Filter Significant DEGs
↓
Obtain Protein Sequences
↓
BLAST against UniProt
↓
Download UniProt Annotation
↓
Merge and Interpret Results

3 Step 1: Inspect the DEG Dataset

Check the identifiers in your DEG file.

GeneID	Gene Symbol	log2FC	pvalue	padj
ENSG00000141510	TP53	2.31	0.0001	0.001
ENSG00000146648	EGFR	1.74	0.0005	0.006
ENSG00000171862	PTEN	-1.82	0.0002	0.003

Important: Determine which type of identifiers you have:

- Ensembl (ENSG...)
- Entrez (e.g., 7157)
- Gene symbol (TP53)
- RefSeq (e.g., NM_000546)
- UniProt (e.g., P04637)

4 Why Do We Need Annotation?

DEG analysis gives us a list of gene IDs, but they do not tell us much about their biological roles.

Annotation converts identifiers to biological information

- Gene name
- Function
- Protein product
- Pathways
- Cellular location
- Disease association

DEG analysis tells us which genes changed, while annotation tells us what those genes are and what they do.

5 Major Annotation Databases

Ensembl: Gene & transcript info, Genomic coordinates, Gene identifiers
NCBI: Gene information, Gene IDs, RefSeq, Genomic information
UniProt: Protein information, Protein function, Protein domains, Subcellular localization
Gene Ontology: Biological functions, Molecular functions, Cellular components
KEGG: Biological pathways
Reactome: Biological pathways

6 Step 2: Filter Significant DEGs

Significance criteria:

- padj < 0.05
- |log2FoldChange| ≥ 1 (optional)

Separate the genes into:

UP-regulated DEGs (red arrow up) / DOWN-regulated DEGs (blue arrow down)

Why?
To focus downstream analysis on biologically significant genes.

7 Step 3: Identify the Corresponding Protein

A gene is not directly submitted to protein BLAST. We need to map:

Gene (ENSG00000141510)
↓
Transcript (ENST...)
↓
Protein (ENSP...)
↓
Protein sequence (FASTA)

Example:
ENSG00000141510
↓
TP53
↓
TP53 protein
↓
Protein FASTA

8 Step 4: Obtain Protein FASTA

Possible sources:

- Ensembl
- NCBI
- UniProt
- GENCODE

Output: DEG_proteins.fasta

9 What is FASTA?

FASTA is a standard format for biological sequences.

>Protein_ID
MEEQSPISVPEPLSQETFSDLWKLLPENNV...

Header (>Protein_ID) Sequence (amino acid sequence)

10 Step 5: Why BLAST?

BLAST = Basic Local Alignment Search Tool
Compares your protein sequence against known sequences.

Your protein (FASTA) → BLASTP → Reference protein database (UniProt)

Purpose: To identify similar known proteins.

11 BLASTP vs BLASTN

BLAST type	Query	Used for
BLASTN	DNA/RNA	Nucleotide comparison
BLASTP	Protein	Protein comparison
BLASTX	DNA → protein	Translation-based search
TBLASTN	Protein → DNA	Protein against translated DNA

For our workflow:
Protein FASTA → BLASTP

12 Step 6: BLAST Against UniProt

Use the UniProt BLAST service.

Input: DEG_proteins.fasta
Search against UniProt protein database
Output provides:
• UniProt accession
• Protein name
• Organism
• Sequence identity
• Query coverage
• E-value
• Alignment

13 Understanding BLAST Results

Important parameters:

Parameter	Value
Percentage Identity	99% → very high similarity 70% → substantial similarity 30% → careful interpretation
Query Coverage	100% → entire sequence 50% → half of sequence
E-value	Lower E-value = stronger evidence (e.g., 0, 1e-50)

14 How to Select the Best BLAST Hit

Consider multiple criteria:

- Percentage identity
- Query coverage
- E-value
- Protein length
- Organism (e.g., Homo sapiens)
- Protein name (relevant function)
- Reviewed status
- Conserved domains

Ideal example:
Identity = 98%
Coverage = 100%
E-value = 0
Organism = Homo sapiens
Reviewed = Yes

15 Reviewed vs Unreviewed UniProt

Swiss-Prot (Reviewed)
• Manually curated
• Higher-quality annotation
• Literature-supported information

TrEMBL (Unreviewed)
• Computationally annotated
• May not have extensive manual curation

Recommendation:
Prefer a reviewed UniProtKB/Swiss-Prot entry when available.

16 Step 7: Record the Best BLAST Hit

Gene	UniProt	Identity	Coverage	E-value	Reviewed
TP53	P04637	99.2%	100%	0	Yes
EGFR	P00533	100%	100%	0	Yes
PTEN	P60484	98.5%	99%	1e-50	Yes
...	...	...	...	...	...

Save as: BLAST_results.tsv

17 Step 8: Obtain UniProt Annotation

Use the UniProt accession obtained from BLAST.

UniProt accession: P04637, P00533, P60484
Search in UniProt and download TSV file
Output: UniProt_annotation.tsv

18 What Information Can Be Downloaded?

Basic annotation	Functional annotation	Cross references
• UniProt accession • Protein name • Gene name • Organism • Protein length	• Function • Catalytic activity • Subcellular location • Pathway	• Ensembl • NCBI Gene • RefSeq • KEGG • PDB

19 Gene Ontology Information

Biological Process (BP)	Molecular Function (MF)	Cellular Component (CC)
• DNA repair • Cell cycle • Signal transduction • ...	• DNA binding • ATP binding • Kinase activity • ...	• Nucleus • Cytoplasm • Plasma membrane • ...

20 Protein Domain Annotation

InterPro: Protein families and domains
Pfam: Protein domains
PROSITE: Functional sequence patterns/sites

Helps answer:
What structural or functional regions does this protein contain?

21 Step 9: Download UniProt TSV

Recommended fields:

- Entry
- Entry Name
- Reviewed
- Protein names
- Gene Names
- Organism
- Length
- Function
- Subcellular location
- Pathway
- GO Biological Process
- GO Molecular Function
- GO Cellular Component
- InterPro
- Pfam
- Ensembl
- RefSeq

TSV

22 Step 10: Merge the Two Files

File 1: BLAST_results.tsv
• GeneID
• Gene_symbol
• UniProt_Accession
• Identity
• Coverage
• E-value
• ...

File 2: UniProt_annotation.tsv
• Entry
• Protein name
• GO
• InterPro
• KEGG
• Function
• ...

+

23 Merge Key

Common identifier:
BLAST_results UniProt_Accession → UniProt annotation Entry

Use this as the merge key.
UniProt_Accession = Entry

24 Final Annotated DEG Table

Gene	log2FC	padj	UniProt	Protein	GO-BP	GO-MF	GO-CC	KEGG
TP53	2.31	0.001	P04637	p53	DNA repair	DNA binding	Nucleus	hsa04115
EGFR	1.74	0.006	P00533	EGFR	signaling	kinase activity	Membrane	hsa04012
PTEN	-1.82	0.003	P60484	PTEN	cell cycle	ATP binding	Cytoplasm	hsa04110
...	...	...	...	...	...	...	...	...

22 Step 10: Merge the Two Files

Recommended fields:

- Entry
- Entry Name
- Reviewed
- Protein names
- Gene Names
- Organism
- Length
- Function
- Subcellular location
- Pathway
- GO Biological Process
- GO Molecular Function
- GO Cellular Component
- InterPro
- Pfam
- Ensembl
- RefSeq

Output: UniProt_annotation.tsv

23 Merge Key

Common identifier:
BLAST_results UniProt_Accession → UniProt annotation Entry

Use this as the merge key.
UniProt_Accession = Entry

24 Final Annotated DEG Table

Gene	log2FC	padj	UniProt	Protein	GO-BP	GO-MF	GO-CC	KEGG
TP53	2.31	0.001	P04637	p53	DNA repair	DNA binding	Nucleus	hsa04115
EGFR	1.74	0.006	P00533	EGFR	signaling	kinase activity	Membrane	hsa04012
PTEN	-1.82	0.003	P60484	PTEN	cell cycle	ATP binding	Cytoplasm	hsa04110
...	...	...	...	...	...	...	...	...

This becomes your Master DEG Annotation Table.

25 Why Keep DEG Statistics?

Do not remove:

- log2FoldChange
- p-value
- adjusted p-value
- baseMean
- regulation

Because later we can connect:
Differential expression + Protein annotation + Functional annotation → Biological interpretation

26 From Data to Biological Insights

Differential Expression (DEGs)
↓
Gene Annotation (UniProt)
↓
GO / KEGG / Pathways
↓
Biological Interpretation

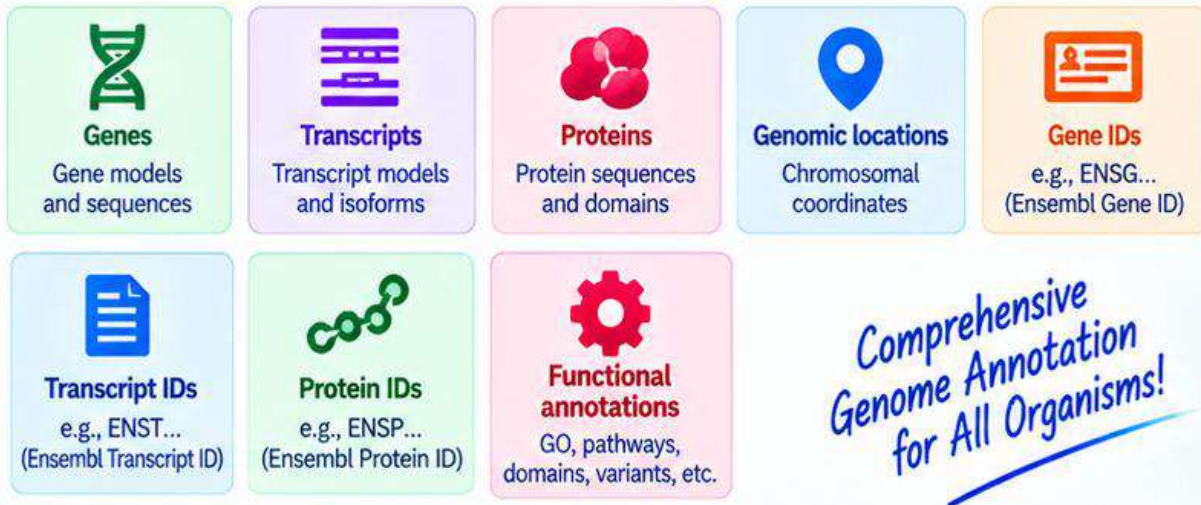
Genes Tell a Story When We Decode Them!

1 Ensembl: Basic Concept

e!Ensembl
Genome Information
for a Better Tomorrow

What is Ensembl?

Ensembl is a **genome annotation resource** that provides information about:



*Comprehensive
Genome Annotation
for All Organisms!*

Main purpose

To establish the relationship:



Example: TP53

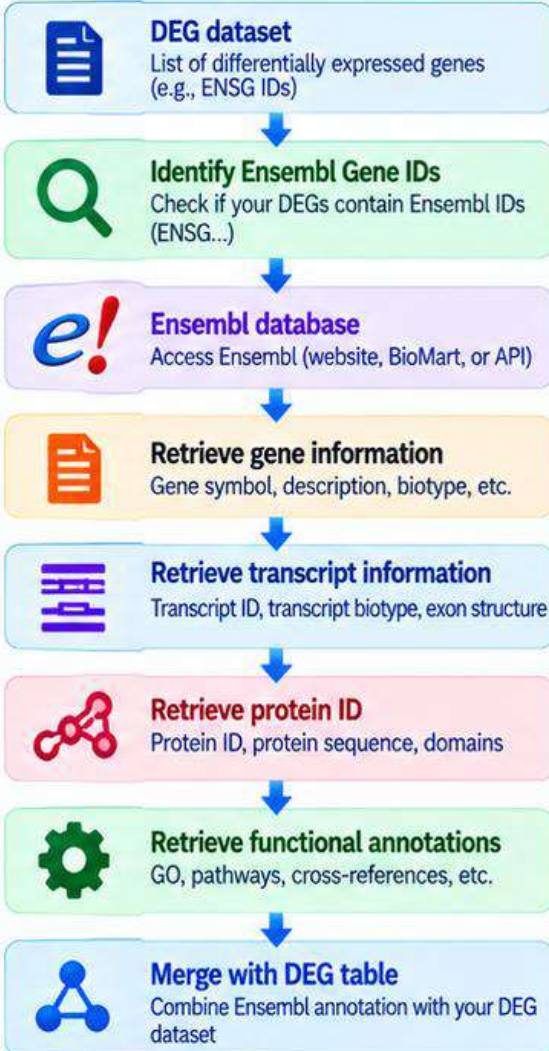


Ensembl links genomic information to biological function.
From sequence to function — Ensembl provides the complete picture!

2 Ensembl: Methodology

Explore Genomes
Understand Biology
Enable Discovery

Workflow



Output: Information Retrieved from Ensembl

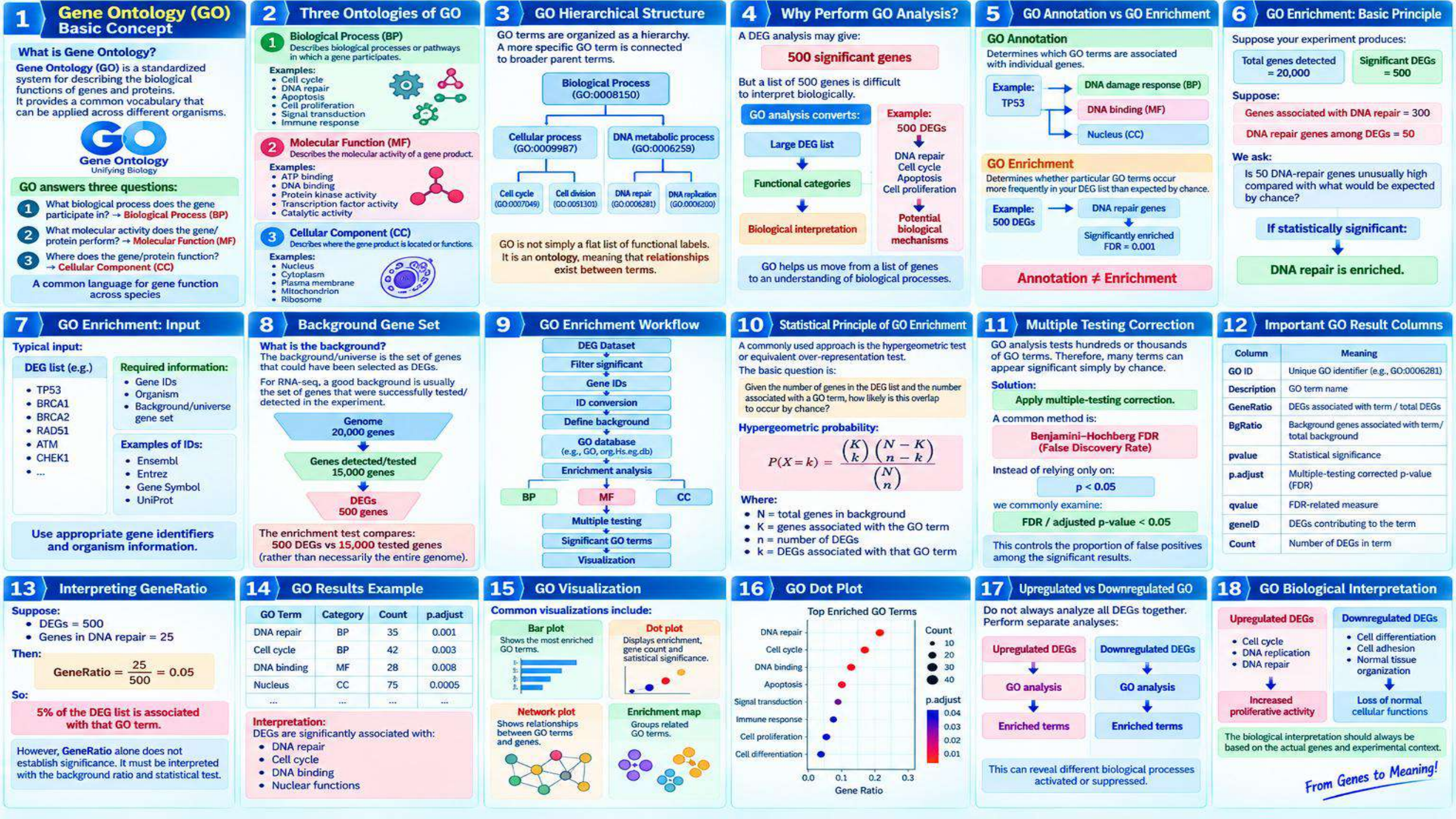
	Ensembl Gene ID	(e.g., ENSG00000141510)
	Gene symbol	(e.g., TP53)
	Transcript ID	(e.g., ENST00000269305)
	Protein ID	(e.g., ENSP00000269305)
	Gene description	(e.g., tumor protein p53)
	Chromosomal location	(e.g., 17p13.1)
	GO annotations	Biological Process (BP) Molecular Function (MF) Cellular Component (CC)
	Pathway information	(e.g., KEGG, Reactome)
	Cross-references	(e.g., UniProt, RefSeq, NCBI Gene, HGNC, PDB, etc.)

From Genome to Function

Ensembl provides the genomic, transcriptomic, proteomic and functional context in a single integrated resource.



*Open Data
for a Healthier
World*



1 KEGG: Basic Concept

What is KEGG?



Kyoto Encyclopedia of Genes and Genomes

KEGG is a database/resource for understanding biological systems through:

- Genes
- Proteins
- Metabolic pathways
- Signaling pathways
- Cellular processes
- Diseases
- Organism-specific pathways

Main question

"Which biological pathways are represented or enriched among my DEGs?"

2 GO vs KEGG

Gene Ontology (GO)

Functional ontology

BP / MF / CC

Describes gene functions

Functional categories

Example: DNA binding

KEGG

Pathway-oriented resource

Signaling / metabolic pathways

Connects genes into pathways

Biological system-level interpretation

Example: MAPK signaling

Simple distinction

GO = What does the gene do?

KEGG = Which pathway/system does the gene participate in?

GO
Gene functions



KEGG
Biological pathways

Complementary, not competitive!

3 KEGG Organization

KEGG contains several types of information.

KEGG Orthology (KO)

Groups genes with similar functions across organisms.



Pathways

Examples:

- MAPK signaling
- PI3K-Akt signaling
- Cell cycle
- Apoptosis
- Metabolic pathways



Disease pathways

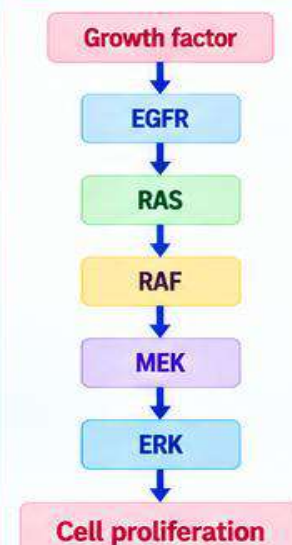
Can connect molecular changes with disease-associated mechanisms.



KEGG links genes to pathways, pathways to diseases, and molecular data to biology.

4 KEGG Pathway Structure

Example: MAPK signaling pathway (simplified)



- A pathway provides relationships among genes/proteins, rather than merely listing their functions.
- Pathways show how molecular events are connected.
- KEGG pathway maps can help interpret the biological context of DEGs.
- Many pathways include signaling, metabolic and regulatory components.

5 KEGG Analysis Input

Typical input:

Significant DEGs

Required information:

- Gene IDs
- Organism
- Background/universe gene set

Example:

TP53
EGFR
AKT1
PIK3CA
KRAS
MYC
...

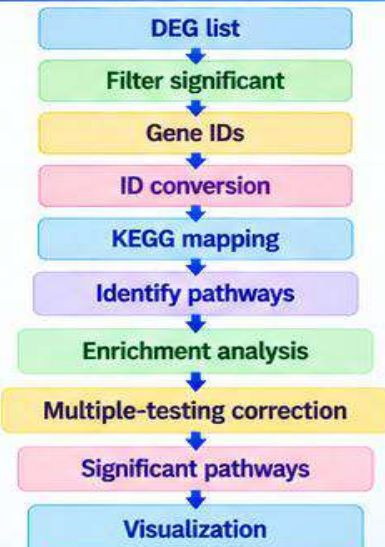
Examples of IDs:

- Ensembl
- Entrez
- Gene Symbol
- UniProt

The genes need to be converted to an identifier recognized by the selected KEGG organism/database. For human analysis, the organism-specific KEGG identifier is:

hsa

6 KEGG Enrichment Workflow



7 KEGG Enrichment: Basic Principle

Suppose:

- Total background genes = 15,000
- DEGs = 500

A KEGG pathway contains:

- 100 genes

Among your DEGs:

- 30 genes

The question is:

- Is 30 genes an unusually large representation of this pathway?

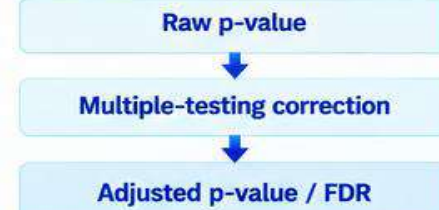
If yes:

The pathway is significantly enriched.

8 KEGG Statistical Analysis

KEGG over-representation analysis can use the same general statistical framework as GO enrichment.

Hypergeometric / Fisher's exact test

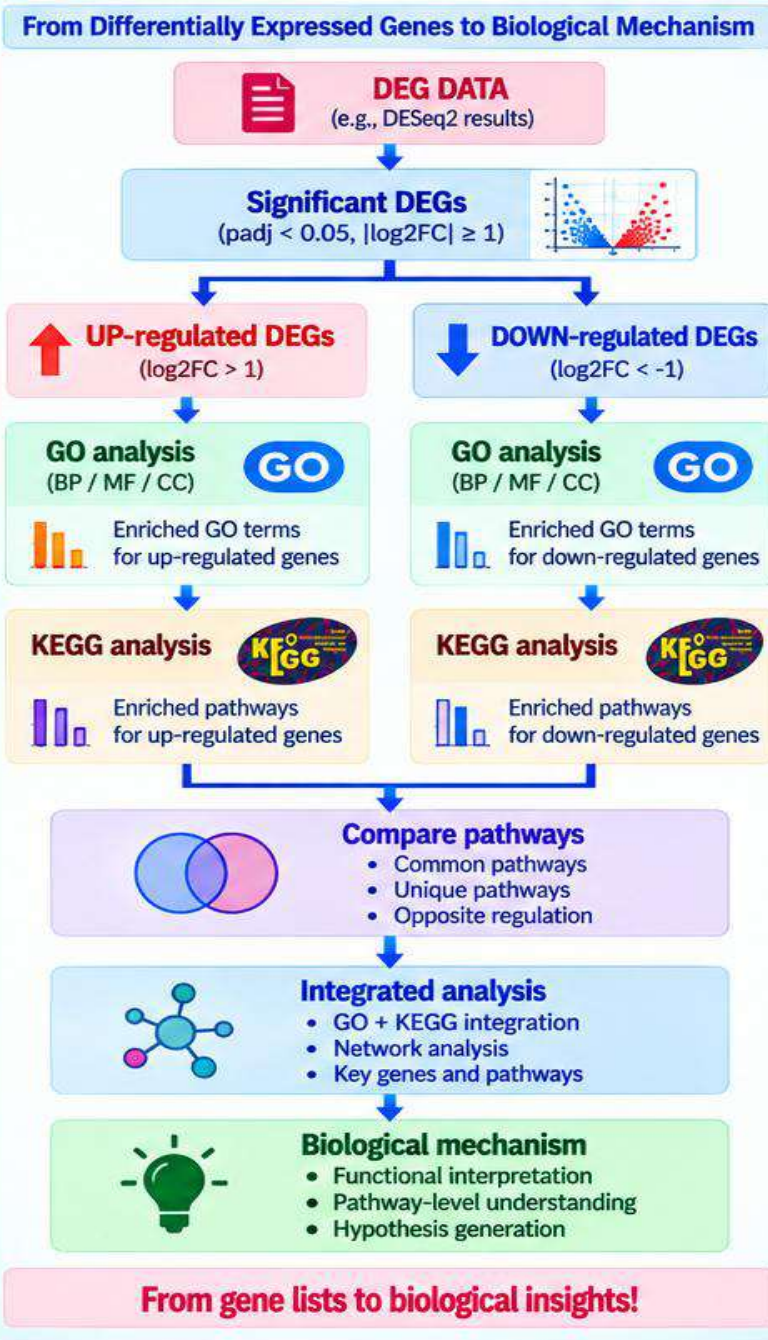


A commonly used threshold is:

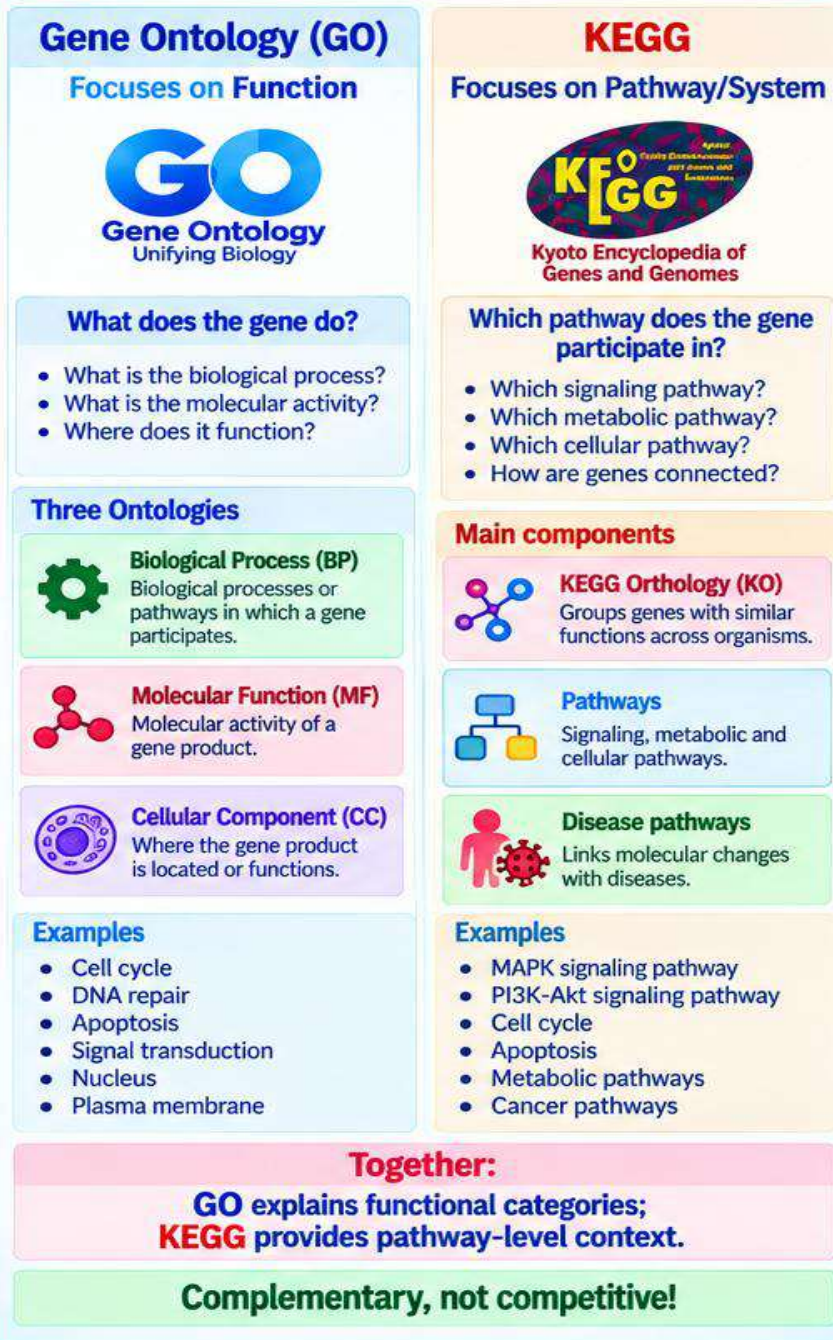
FDR < 0.05

The precise threshold should be defined before interpretation and reported with the analysis.

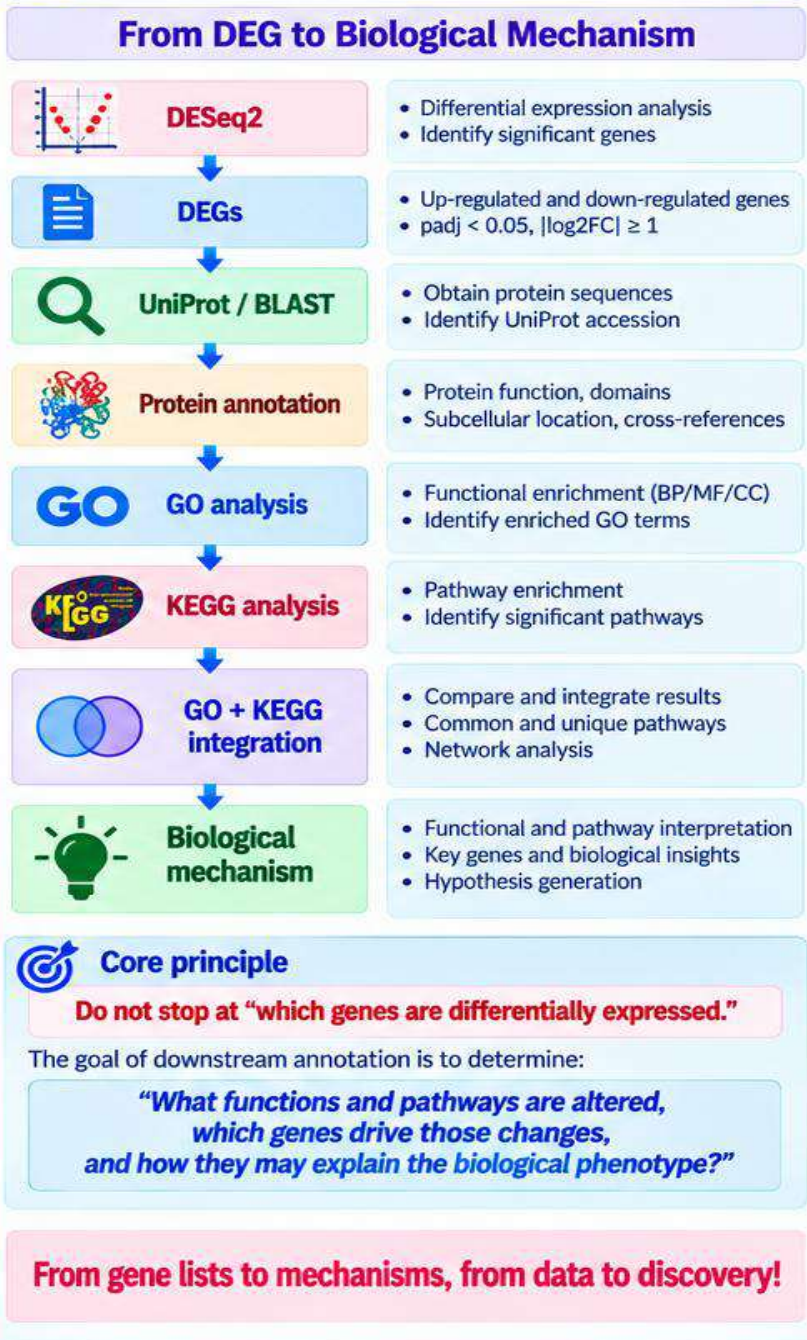
38 Final GO + KEGG Workflow



39 Key Difference: GO vs KEGG



40 Final Take-Home Message



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